

CIVICCONNECT — 4-MEMBER TEAM DIVISIONS

Technical Architecture, Module Ownership & Individual Contribution Matrix

This specification establishes a clean, non-overlapping architectural division of the CivicConnect Municipal Grievance & Triage Platform across 4 specialized engineering roles. Each division encapsulates a complete subsystem with clear code ownership, architectural responsibilities, and presentation deliverables.

EXECUTIVE DIVISION OVERVIEW

Division 1

AI, NLP & Intelligent Triage Lead

Artificial Intelligence & Vision Pipeline

Division 2

Backend API, Database & Security Lead

Cloud Infrastructure, PostgreSQL & Auth

Division 3

Citizen Portal & Public GIS Lead

Frontend UI/UX, Spatial Maps & Tracking

Division 4

Department Ops & Field Mobility Lead

Operations Center, Field Mobile & E2E Testing

SYSTEM INTERACTION & DATA FLOW

1. Division 3 captures citizen complaint, geocodes coordinates, and submits to Division 2 API.
2. Division 2 validates stateless Firebase JWT, writes complaint to Supabase PostgreSQL, and triggers Division 1 AI pipeline.
3. Division 1 evaluates Gemini 2.5 Flash / NLP classifier, assigns confidence, calculates SLA priority, and routes to department.
4. Division 4 Department Coordinator assigns field engineer; engineer receives real-time mobile notification, confirms GPS on-site arrival, uploads Before/After evidence, and manager approves resolution.

DIVISION 1: AI, NLP & INTELLIGENT TRIAGE LEAD

Core Specialization: Machine Learning, NLP, Multimodal Vision & Triage Algorithms

- Multimodal Vision & NLP: Integrated Google Gemini 2.5 Flash for simultaneous image recognition and text semantics.
- Zero-Latency Local Fallback: Built local Naive Bayes / TF-IDF classifier (336 training patterns across 8 departments).
- Automated Priority Engine: Deterministic rule engine for instant safety hazard escalation (<5ms response time).
- Confidence-Based Routing: Automated auto-assignment for confidence >= 70%; admin review flag for < 70%.
- AI Configuration Dashboard: Admin portal to adjust model temperatures, view confusion matrices, and audit logs.

Code Files Owned (Division 1):

functions/ai.js, functions/nlpClassifier.js, functions/priority.js, functions/departmentRoutingService.js, src/pages/admin/AIConfig.jsx

DIVISION 2: BACKEND API, DATABASE & SECURITY LEAD

Core Specialization: Cloud Infrastructure, PostgreSQL Schema, JWT Auth & Security

- Normalized Database Schema: Engineered Supabase PostgreSQL schema with primary/foreign keys and B-Tree indexes.
- Stateless JWT Authentication: Bridged Firebase Auth tokens with Supabase PostgreSQL user roles and permissions.
- High-Throughput REST API: Engineered modular Express routers with fast-path ingest (<100ms response time).
- Immutable Audit Trail: Automated audit logging on every complaint status change with actor ID, role, and timestamps.
- Connection Pooling & Scaling: Configured Supervisor connection pooling for handling 10,000+ pooled connections.

Code Files Owned (Division 2):

functions/db.js, functions/authMiddleware.js, functions/lib/supabaseAdmin.js, server/index.js, functions/routes/issues.js, functions/routes/departments.js

DIVISION 3: CITIZEN EXPERIENCE & PUBLIC GIS LEAD

Core Specialization: Frontend UI/UX, Spatial GIS Heatmaps & Citizen Workflows

- Multi-Step Grievance Wizard: Built responsive reporting form with GPS coordinate geocoding and image capture.
- Interactive GIS Public Heatmap: Developed real-time Leaflet/MapLibre map with category filters and spatial clustering.
- Zero-Auth Public Tracking: Created public status lookup portal by Reference ID (CC-YYYY-XXXXXX).
- Glassmorphism Design System: Built accessible UI tokens, responsive dark/light theme, and micro-interactions.
- Citizen Resolution Feedback: Implemented satisfaction rating (1-5 stars) and repair verification/reopen workflow.

Code Files Owned (Division 3):

src/pages/citizen/ReportIssue.jsx, src/pages/citizen/Dashboard.jsx, src/pages/public/PublicMap.jsx, src/pages/public/PublicTrack.jsx, src/context/AuthContext.jsx

DIVISION 4: DEPARTMENT OPS & FIELD MOBILITY LEAD

Core Specialization: Department Workflows, Field Mobility, Real-Time State Machine & E2E

- Strict Department Isolation: Built Coordinator dashboard with 403 enforcement preventing cross-department leaks.
- Field Engineer Mobile Interface: Mobile-first portal for task inspection, route navigation, and work submission.
- GPS Check-in & Evidence Proof: Built GPS verification confirming physical on-site presence before repair starts.
- Canonical State Machine: Enforced ASSIGNED -> ACCEPTED -> EN_ROUTE -> ON_SITE -> IN_PROGRESS -> CLOSED state machine.
- Automated Testing Suites: Created 8-department E2E verification test and high-concurrency traffic load test engine.

Code Files Owned (Division 4):

src/pages/department/Dashboard.jsx, src/pages/engineer/Dashboard.jsx, src/services/api/engineerApi.js, test_all_departments_e2e.cjs, load_test_suite.cjs

Ø<ß INDIVIDUAL VIVA & EVALUATION TALKING POINTS

Member 1 — AI & Triage Lead

"I developed the intelligent triage engine. I integrated Google Gemini 2.5 Flash for multimodal photo/text analysis, combined with a local Naive Bayes classifier that guarantees zero-latency fallback routing during network disruptions. I also built the deterministic priority algorithm that escalates life hazards instantly."

Member 2 — Backend & Security Lead

"I designed the cloud backend architecture, database schema, and security framework. I engineered the normalized PostgreSQL schema on Supabase, implemented connection pooling to handle high concurrency, and created the stateless JWT authentication layer that enforces Role-Based Access Control."

Member 3 — Citizen Portal & GIS Lead

"I led the citizen-facing web applications, public transparency portal, and spatial GIS integration. I developed the multi-step grievance filing wizard with live GPS geolocation, designed the interactive public heatmap for community awareness, and built the zero-auth public tracking system."

Member 4 — Department Ops & Field Lead

"I developed the department management portal, field engineer mobility system, and real-time lifecycle orchestrator. I enforced strict cross-department data isolation, built the mobile-first field engineer workflow with GPS arrival verification, and created the automated 8-department E2E and load test suites."